Letters

A Novel LCL Resonant Converter With Inherent CC-CV Output for On-Board Chargers of Plug-In Electric Vehicles

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***Abstract*—The *LLC* Resonant converter is widely used in on-board charger applications for its high efficiency. However, conventional *LLC* converter with pulse-frequency-modulation suffers from a wide switching frequency range for realizing constant current (CC) and constant voltage (CV) charges. To address this problem, different from the traditional solution, a novel *LCL* resonant converter operating at a fixed resonant frequency with inherent CC and CV output characteristics is proposed in this letter. In the proposed concept, the CC and CV charges are accomplished by the special *LCL* tank. With the increasing of battery voltage during charging process, the resonant tank of the proposed converter is equal to the conventional *LCL* tank and *LC* tank for CC mode and CV mode, respectively. Thus, the sensing and control circuits are canceled while the wide zero-voltage-switching operation for primary switches and zero-current-switching for secondary diodes still be maintained. Therefore, the fixed resonant frequency operation, high efficiency, high reliability, and low cost are key features of the proposed converter. A 1-kW prototype with 400 V input and 250–400 V output is built and tested. The full load efficiency of 96.7% and higher than 96% for the CC mode were both achieved. The experimental results demonstrate the advance and validity of the proposed converter.**

***Index Terms*—Constant current/constant voltage (CC/CV) charge, fixed-frequency operation, high efficiency, *LCL* resonant converter, reconfiguration rectifier circuit.**

I. Introduction

PLUG-IN electric vehicles (PEVs) are becoming increasingly popular due to their energy efficiency and environmental friendliness [1]. In PEVs, the on-board charger (OBC) is the key component of the PEVs [2]. Owing to the wide battery voltage range, the typical architecture of the OBC is the front-end PFC converter cascaded by an isolated dc/dc converter. The front-end PFC converter is used to realize high power factor and low-input current harmonics. The post-stage isolated dc/dc converter is employed to carry out the desired constant current (CC) and constant voltage (CV) charging for the battery with a wide voltage range (typically 250–400 V, as shown in Fig. 1 [3]).

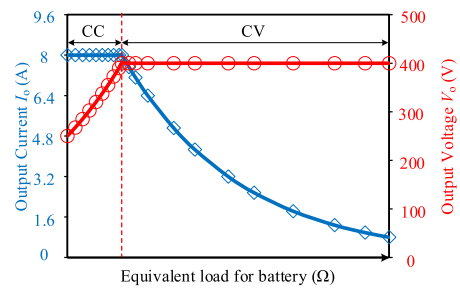

Fig. 1. Charging profile of a high voltage li-ion battery [3].

To obtain the high conversion efficiency and the desired charging profiles of the isolated dc/dc converter, a series of isolated dc-dc converters have been investigated, which can be classified into the following categories.

The first category is to employ the PWM type dc/dc converter, where the phase-shift full-bridge (PSFB) converter is widely utilized topology because of its fixed-frequency operation, and zero-voltage-switching (ZVS) for primary switches. However, the limited ZVS range and duty loss are the main shortcomings [4].

The second category is to employ the resonant type dc/dc converters. Among the resonant type topologies, the *LLC* resonant converter with pulse frequency modulation (PFM) becomes a popular candidate because it allows ZVS switching for the primary switches and zero-current switching (ZCS) for secondary diodes [3], [5]. However, a main drawback of the *LLC* converter employing PFM modulation is its wide switching frequency range, which decreases the efficiency and increased difficulty in designing magnetic components and EMI filters. In order to overcome this issue, a dual full bridge *LLC* converter [6] is proposed to obtain the fixed-frequency operation and high

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efficiency. However, the light load efficiency is significantly reduced by the low component utilization. Furthermore, an additional buck converter [7] is inserted after the *LLC* converter to implement the required CC-CV charging characteristics. In this design, the *LLC* converter serves as a dc transformer for achieving the highest efficiency during the entire charging process. However, the total efficiency is decreased because of the increased number of power conversion stage.

Another candidate is hybrid type converters. One hybrid converter is presented by incorporating PSFB converter and *LLC* converter [8]. Thus, the wide ZVS range, fixed frequency operation and high efficiency are obtained simultaneously. However, this converter suffers from the drawbacks including the ZVS loss of the lagging leg and increased turn-OFF current for the leading leg.

In summary, the CC-CV charging characteristics of the aforementioned OBC topologies are implemented by employing suitable closed-loop control with the battery voltage and current sense. As compared to the control strategy-based CC-CV implementation method, the CC-CV charging can be achieved using special topologies with natural CC-CV characteristics, which is another completely different solution. In this solution, the sensing and control circuits are canceled while the wide ZVS operation for primary switches can still be maintained [9], [10]. As a result, the fixed resonant frequency operation, high efficiency, high reliability, and low cost are its key features. An *LCL*-T resonant converter [9] with clamp diodes is developed. Unfortunately, the *LCL*-T converter has the natural CC output. However, the CV charging is carried out by the clamp diodes. Hence, the efficiency of the CV stage is decreased because of the circulating current flowing back to the input source through clamp diodes. Similarly, a CLCL resonant converter [10] with CC-CV characteristics is presented. However, it is worth noting that the topology employed two full-bridge diode rectifiers, resulting in the low utilization of power devices.

To derive a high efficiency and cost-effective topology with inherent CC-CV characteristics, a novel *LCL* resonant topology is proposed in this letter. The circuit configuration, theoretical analysis, and design considerations of the proposed converter are introduced in this letter. A 1-kW prototype with 400 V input and 250–400 V output is built and tested. The experimental results demonstrate the advance and validity of the proposed converter. An efficiency higher than 96% for CC mode and 96.7% under full load condition were both achieved.

II. Proposed Converter and Theoretical Analysis

A. Circuit Configuration

The proposed converter is depicted in Fig. 2. Same to the methods in [11], [12], [13], and [14], the battery is modeled as an equivalent resistance R_{BAT}. The converter operates at the fixed resonant frequency. MOSFETs Q_1 and Q_2 (Q_3 and Q_4) operate complementarily with the fixed duty cycle of 0.5 and the deadtime t_d. The diagonal switches, i.e., Q_1 and Q_4 (Q_2 and Q_3) are turned on and off simultaneously. As shown in Fig. 2, the novel *LCL* resonant tank composed of resonant

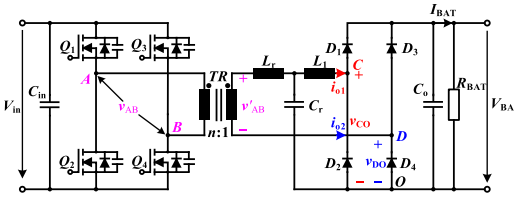

Fig. 2. Schematic of the proposed converter.

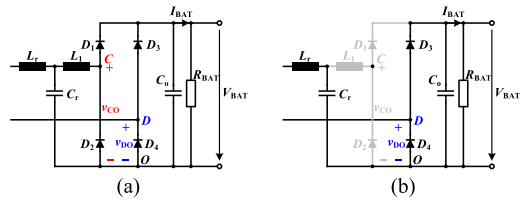

Fig. 3. Operation modes of the secondary side resonant tank and rectifier circuit. (a) CC mode. (b) CV mode.

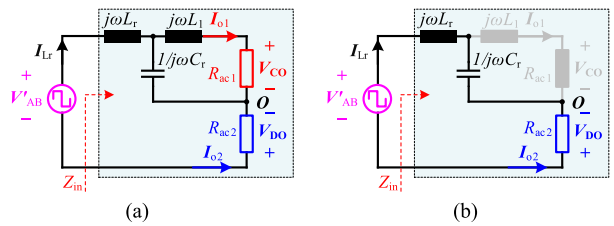

Fig. 4. AC equivalent circuits of the proposed converter in different operation modes. (a) CC mode. (b) CV mode.

inductor L_{r} and L_1 and resonant capacitor C_{r} is employed for achieving the natural CC-CV characteristics. At the beginning of CC charging stage, the proposed converter can be equivalent to an *LCL* resonant converter for its CC output characteristic. With the increasing of the equivalent resistance R_{BAT}, the battery voltage V_{BAT} is gradually increased. At the transition point for CC and CV mode, the battery voltage will be higher than the voltage across resonant capacitor C_r. The diodes D_1 and D_2 are reverse biased. The resonant tank for the proposed converter will smoothly transfer into an *LC* resonant circuit with CV output characteristics. The charging process enters into CV stage. The corresponding equivalent circuit of the secondary side for the CC and CV mode are depicted in Fig. 3(a) and (b), respectively. The inherent CC-CV characteristics of the proposed converter are detailly analyzed and introduced in Section II-B.

B. Theoretical Analysis

Using the fundamental harmonic approximation method, the equivalent ac circuits of Fig. 3 can be derived, as shown in Fig. 4(a) and (b), respectively. In the following analysis, $\boldsymbol{V}'_{AB}$ and $\boldsymbol{V}'_{AB}$ are adopted to represent the vector and the root-mean-square of $\boldsymbol{v}'_{AB}$, respectively, and other variables are denoted in similarly.

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According to Fig. 4(a), the following equations can be obtained:

$$\begin{cases} \boldsymbol{V}'_{\mathrm{AB}} = -\boldsymbol{I}_{o2}\left(j\omega L_r + R_{\mathrm{ac2}}\right) - \frac{1}{j\omega C_r}\left(\boldsymbol{I}_{o1} + \boldsymbol{I}_{o2}\right) \\ 0 = \boldsymbol{I}_{o1}\left(j\omega L_1 + R_{\mathrm{ac1}}\right) + \frac{1}{j\omega C_r}\left(\boldsymbol{I}_{o1} + \boldsymbol{I}_{o2}\right) \end{cases} \tag{1}$$

where $\boldsymbol{V}'_{AB} = \boldsymbol{V}_{\mathrm{AB}}/n, \omega = 2\pi f_{\mathrm{s}}$ is the switching angle frequency, R_{ac1} is the ac equivalent resistance between terminal C and O. Similarly, R_{ac2} is the ac equivalent resistance between terminal D and O. Due to the parallel connection of the two half-bridge rectifier circuits, the relationship between R_{ac1}, R_{ac2}, and R_{BAT} can be given as

$$R_{\mathrm{ac1}} \parallel R_{\mathrm{ac2}} = \frac{2}{\pi^2} R_{\mathrm{BAT}}. \tag{2}$$

For simplicity of the analysis, following assumptions are made:

$$\begin{cases} L_r = L_1 \\ j\omega L_r = j\omega L_1 = -\frac{1}{j\omega C_r} \end{cases}. \tag{3}$$

Substituting (3) into (1), the following equations can be derived:

$$\begin{cases} \boldsymbol{V}'_{\mathrm{AB}} = \boldsymbol{I}_{Lr}\left(\omega C_r R_{\mathrm{ac1}} R_{\mathrm{ac2}} + \omega L_1\right)/\omega C_r R_{\mathrm{ac1}} \\ \boldsymbol{I}_{o1} = \boldsymbol{I}_{Lr}/j\omega C_r R_{\mathrm{ac1}} \\ \boldsymbol{I}_{o2} = -\boldsymbol{I}_{Lr} \\ \boldsymbol{V}_{\mathrm{CO}} = \boldsymbol{I}_{Lr}/j\omega C_r \\ \boldsymbol{V}_{\mathrm{DO}} = -\boldsymbol{I}_{Lr} R_{\mathrm{ac2}} \end{cases}. \tag{4}$$

As the two half-rectifier circuits are connected in parallel, the output voltage V_{BAT} will be determined by the higher value of V_{CO} and V_{DO}. According to (4), the value of V_{CO} and V_{DO} can be given as follows

$$\begin{cases} V_{\mathrm{CO}} = I_{Lr}/\omega C_r \\ V_{\mathrm{DO}} = -I_{Lr} R_{\mathrm{ac2}} \end{cases}. \tag{5}$$

1) CC Charging Stage: At the beginning of CC charging stage, R_{BAT} is relatively small. The proposed converter can be regard as an *LCL* resonant converter and the CC output is realized. By (5), the voltage V_{DO} is smaller than the voltage V_{CO}. Thus, V_{DO} is equal to V_{CO} and $R_{\mathrm{ac2}} = 1/\omega C_{\mathrm{r}}$. The output current I_{BAT} is calculated as

$$I_{\mathrm{BAT}} = \frac{\sqrt{2}}{\pi}\left(\frac{V_{\mathrm{CO}}}{R_{\mathrm{ac1}}} + \frac{V_{\mathrm{DO}}}{R_{\mathrm{ac2}}}\right). \tag{6}$$

According to (4), (6) and $R_{\mathrm{ac2}} = 1/\omega C_{\mathrm{r}}$, it is not difficult to derive the charging current I_{BAT} as

$$I_{\mathrm{BAT}} = \frac{\sqrt{2}}{\pi}\omega C_r V'_{\mathrm{AB}} = \frac{4}{n\pi^2}\omega C_r V'_{\mathrm{AB}}. \tag{7}$$

Furthermore, the input impedance Z_{in} can be expressed as

$$Z_{\mathrm{in}} = \frac{\boldsymbol{V}'_{\mathrm{AB}}}{\boldsymbol{I}_{Lr}} = \frac{\omega C_r R_{\mathrm{ac1}} R_{\mathrm{ac2}} + \omega L_1}{\omega C_r R_{\mathrm{ac1}}}. \tag{8}$$

Thus, the input impedance Z_{in} is a pure resistance and an input zero-phase angle (ZPA) can be achieved. As a result, the circulating energy is reduced dramatically.

2) CV Charging Stage: With the increasing of CC charging time, the battery voltage is gradually increased. At the transition point for CC and CV mode, the battery voltage will be higher than the voltage across resonant capacitor C_r. At this time, the output characteristics of the proposed converter will transfer into CV output. As a consequence, the proposed converter will also automatically change to an *LC* resonant converter. The equivalent ac circuit of the proposed converter in the CV mode is shown in Fig. 4(b). Equation (9) can be established:

$$\begin{cases} \boldsymbol{V}'_{\mathrm{AB}} = \boldsymbol{I}_{Lr} R_{\mathrm{ac2}} \\ \boldsymbol{I}_{o2} = -\boldsymbol{I}_{Lr} \\ \boldsymbol{V}_{\mathrm{DO}} = \boldsymbol{I}_{o2} R_{\mathrm{ac2}} \end{cases}. \tag{9}$$

Therefore, the charging voltage V_{BAT}, which is equal to $(\pi/\sqrt{2})V_{DO}$, can be deduced as

$$V_{\mathrm{BAT}} = \frac{\pi}{\sqrt{2}} V'_{\mathrm{AB}} = \frac{2}{n} V_{\mathrm{in}}. \tag{10}$$

Hence, CV charging can be realized. The input impedance Z_{in} is equal to R_{ac2}, which is a pure resistance and equal to $2R_{\mathrm{BAT}}/\pi^2$. Thus, input ZPA can be achieved as well.

3) CC-CV Transition Point: With the increasing of the equivalent resistance R_{BAT}, the battery voltage V_{BAT} is gradually increased. Gradually, the battery will enter into the CV charging stage from CC charging stage. In this letter, the equivalent resistance of the battery at the transition point between CC mode and CV mode is defined as $R_{\mathrm{BAT_CC/CV}}$. At the transition point, the battery voltage will be higher than the voltage across resonant capacitor C_r. Thus, the diodes D_1 and D_2 are reverse biased. Hence, from the resonant tank point of view, the equivalent ac resistance between terminals D and O at the beginning point of CV mode $R_{\mathrm{ac2_CC/CV}}$ should be equal to the value of the equivalent ac resistance at the endpoint of CC mode. Therefore, the equivalent resistance of the battery at the transition point can be deduced as:

$$\begin{cases} R_{\mathrm{ac2_CC/CV}} = \frac{1}{\omega C_r} \\ R_{\mathrm{ac2_CC/CV}} = \frac{2R_{\mathrm{BAT_CC/CV}}}{\pi^2} \end{cases} \Rightarrow R_{\mathrm{BAT_CC/CV}} = \frac{\pi^2}{2\omega C_r} \tag{11}$$

where $R_{\mathrm{BAT_CC/CV}}$ can be obtained from the desired charging curve. Thus, the resonant capacitance C_r can be calculated according to (11). Correspondingly, the resonant inductances of L_r and L_1 can be solved by (3).

Therefore, once the parameters for the resonant tank are solved based on above derived equations, the proposed converter will automatically operate at CC and CV mode with the desired voltage and current specification. Therefore, it is not difficult to know that the proposed converter could achieve inherent CC-CV charging characteristics without additional components and control loops.

III. Design Considerations

The design specifications for the proposed converter are given in Table I. The detailed design considerations are described as follows.

First of all, the key point for parameters design of the proposed converter is the transition point with maximum output power of

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TABLE I
DESIGN SPECIFICATIONS OF THE PROPOSED CONVERTER

Parameter	Value
Input voltage, V_{in}	400 V
Output voltage V_{BAT}, $V_{BAT,min} \sim V_{BAT,max}$	250 ~ 400 V
Maximum output current I_{BAT}	2.5 A
Rated power P_o	1 kW
Deadtime t_d	300 ns
Resonant switching frequency f_s	100 kHz

1 kW. Thus, the equivalent resistance for this point $R_{\mathrm{BAT_CC/CV}}$ can be calculated as

$$R_{\mathrm{BAT_CC/CV}} = \frac{V_{\mathrm{BAT,max}}}{I_{\mathrm{BAT}}} = \frac{400\ \mathrm{V}}{2.5\ \mathrm{A}} = 160\ \Omega. \tag{12}$$

Furthermore, according to (11) and (12), the resonant capacitance C_r is derived as

$$C_r = \frac{\pi^2}{2\omega R_{\mathrm{BAT_CC/CV}}} = \frac{\pi}{4 \times 100 \times 10^3 \times 160} = 49\ \mathrm{nF}. \tag{13}$$

Correspondingly, combining (3) and (13), the resonant inductance L_r and L_1 can be determined by

$$L_r = L_1 = \frac{1}{\omega^2 C_r} = 51.7\ \mu\mathrm{H}. \tag{14}$$

In practical, to get a resonant capacitance close to 49 nF, the capacitance of 47 and 4.7 nF are connected in parallel. Accordingly, the resonant inductance of 49 μH is employed in the prototype.

In addition, the turns-ratio of the transformer *TR* can be calculated as follows

$$V_{\mathrm{BAT,max}} = \frac{2}{n} V_{\mathrm{in}} \Rightarrow n = \frac{2V_{\mathrm{in}}}{V_{\mathrm{BAT,max}}} = 2. \tag{15}$$

Furthermore, thanks to the pure resistance characteristics of the input impedance of the resonant tank Z_{in}, the ZPA operation between the input voltage and resonant current can be ensured, which is very helpful for increasing the efficiency. In order to achieve the ZVS condition for the primary switches, the magnetizing inductance L_m of the transformer *TR* is employed, which is same as the *LLC* resonant converter. The magnetizing inductance of the transformer L_m is calculated as [15]

$$L_m \leq \frac{t_d}{8C_{\mathrm{oss}} f_s} \tag{16}$$

where C_{oss} is the output capacitance of primary switches.

IV. EXPERIMENTAL VERIFICATION

A 1-kW laboratory prototype with the specification given in Table I for high-voltage PEV chargers is built to verify the analysis and effectiveness of the proposed converter. The detailed parameters and the photograph of the prototype are shown in Table II and Fig. 5, respectively.

Same to the methods in [11], [12], [13], and [14], in order to quickly verify the validity of the proposed solution, the battery is modeled as an equivalent resistance R_{BAT} and replaced by the electronic load in the experimental test. The R_{BAT} of 100, 160, and 400 Ω are employed for verifying the performance under the CC mode, transition point, and CV mode condition, respectively.

TABLE II
KEY PARAMETERS OF THE PROTOTYPE

Designators	Part Number/Value
Q_1-Q_4	C3M003090K
TR	Core: PQ3535, L_m= 580μH, n (N_p:N_s)= 2 (20:10)
L_r	Core: PQ3230, 48.43μH
L_1	Core: PQ3230, 45.21μH
C_r	51.7nF (47nF//4.7nF)
D_1-D_4	DSEI120-06A
C_o	390μF/450V

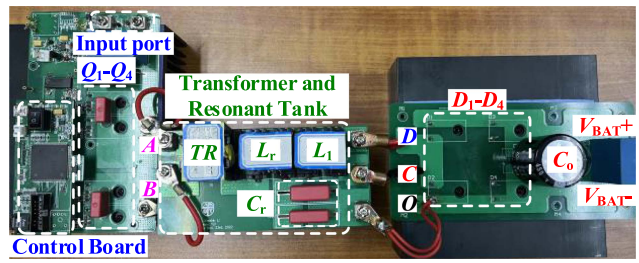

Fig. 5. Photograph of the prototype.

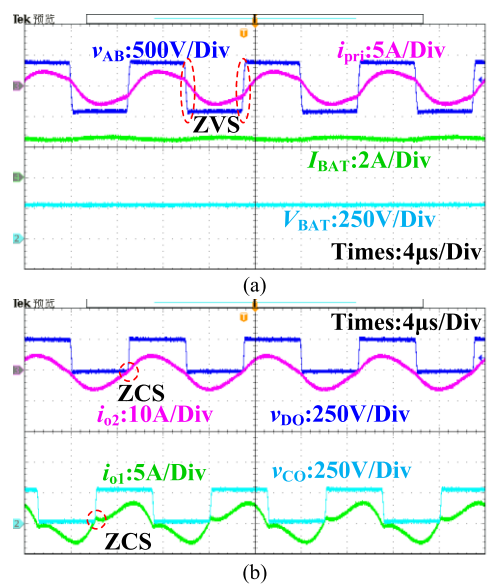

Fig. 6. Experimental waveforms in the CC mode ($R_{\mathrm{BAT}} = 100\ \Omega$). (a) v_{AB}, i_{pri}, I_{BAT}, and V_{BAT}. (b) v_{DO}, i_{o2}, v_{CO}, and i_{o1}.

Fig. 6 shows the experimental waveforms of the proposed converter at the CC mode. As shown in Fig. 6(a), the desired constant output current I_{BAT} of 2.5 A is obtained without any control loop and sensing circuit. Furthermore, the ZVS operation for primary switches can be realized. Fig. 6(b) shows the experimental waveforms of the secondary side current i_{o1}, i_{o2}, and voltage v_{DO}, v_{CO}. It can be found that i_{o1} and i_{o2} are continuous. Thus, the diodes D_1, D_2, D_3, and D_4 are alternately forward biased, constituting the full bridge rectifier. It is not

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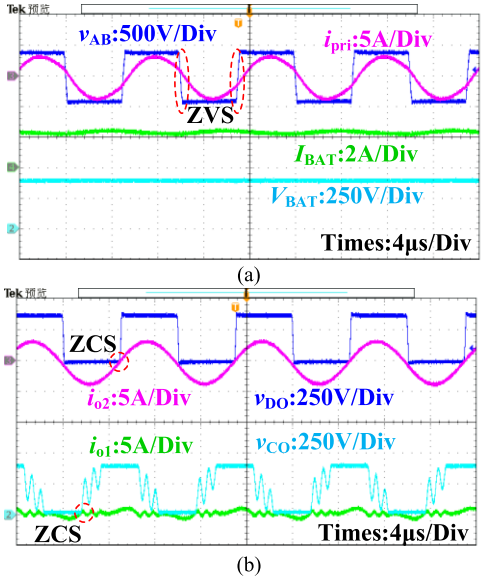

Fig. 7. Experimental waveforms in transition point ($R_{\mathrm{BAT}} = 160\ \Omega$) ($R_{\mathrm{BAT}} = 100\ \Omega$). (a) v_{AB}, i_{pri}, I_{BAT}, and V_{BAT}. (b) v_{DO}, i_{o2}, v_{CO}, and i_{o1}.

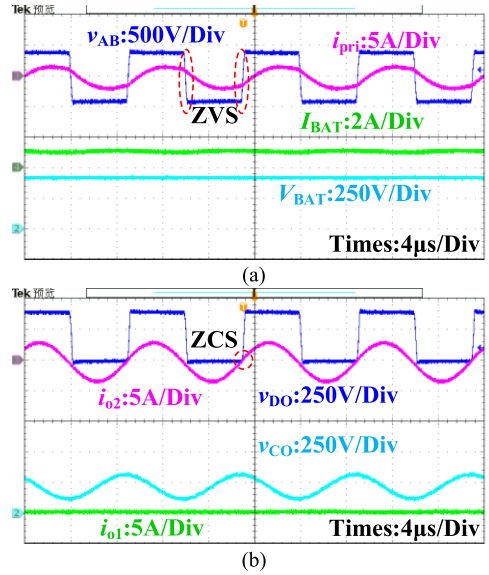

Fig. 8. Experimental waveforms in the CV mode ($R_{\mathrm{BAT}} = 400\ \Omega$). (a) v_{AB}, i_{pri}, I_{BAT}, and V_{BAT}. (b) v_{DO}, i_{o2}, v_{CO}, and i_{o1}.

difficult to conclude that: the desired CC output of 2.5 A is achieved with the contribution of secondary current i_{o1} and i_{o2}. Thus, the proposed converter can be regard as the *LCL* resonant converter. The equivalent operation state of proposed converter is completely consistent with the model shown in Fig. 3(a); and by comparing the voltage v_{CO} (v_{DO}) with the current i_{o1} (i_{o2}), the ZCS switching for all the secondary diodes are realized.

Fig. 7 displays the experimental waveforms in the transition point. Similar to Fig. 6, the ZVS operation for primary switches and ZCS operation for diodes are still realized. Different from Fig. 6(b), the secondary current i_{o1} becomes very small due to the voltage clamping at this transition point. With the increasing of R_{BAT}, the charging process will transfer into CV mode automatically.

The experimental waveforms in the CV mode are indicated in Fig. 8. In this stage, the current level for the secondary current i_{o1} decreases to zero and diodes D_1, D_2 are reverse biased. Thus, the converter is automatically transfer into an *LC* resonant converter and the expected CV of 400 V is realized. The corresponding soft switching feature for primary switches and secondary didoes are same as the CC mode. The equivalent operation state is well matched with the circuit diagram shown in Fig. 3(b).

Moreover, by comparing v_{DO} with i_{o2}, the ZPA condition is implemented during the whole charging process, which greatly reduces the circulation current and improves the efficiency. Meanwhile, the CC-CV output and smooth mode transition are both obtained without sensing circuit and control loops. The experimental results are coincident with the theoretical analysis in Section II.

Fig. 9 depicts the measured charging curve. It is observed that the proposed converter operates at CC mode while $R_{\mathrm{BAT}} < 160\ \Omega$ and at CV mode while $R_{\mathrm{BAT}} > 160\ \Omega$. Obviously, the proposed converter could achieve inherent CC-CV charging

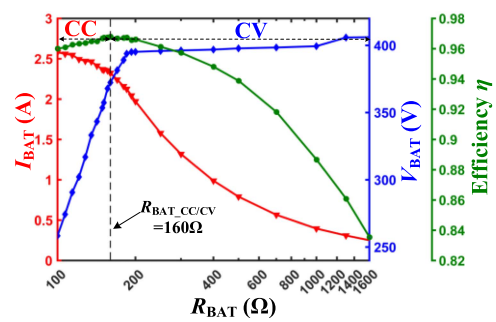

Fig. 9. Measured I_{BAT}, V_{BAT}, and η versus R_{BAT}.

characteristics. Meanwhile, the transition for CC-CV mode is accomplished smoothly without any control, which is consistent well with the theoretical analysis. On the other hand, the charging current and voltage are a little different with the theoretical analysis at the transition point. This is because i_{o1} cannot be changed to zero instantaneously at the transition point. And this is similar to other converters which have inherent CC-CV charging characteristics [9], [10].

In addition, the measured efficiency of the proposed converter is shown in Fig. 9. The maximum efficiency of 96.7% is achieved at full load condition. In the CC mode, the efficiency of the proposed converter is higher than 96%. In the CV mode, when the output power decreases from 903 to 102 W, the efficiency reduces from 96.7% to 83.51%, respectively.

A qualitative comparison between the proposed converter and other converters is given in Table III. As seen, the proposed *LCL* converter can achieve inherent CC-CV output without any control loop and sensing circuit while operating at the resonant frequency. On the other hand, the proposed converter has a higher efficiency than the LLC converter with PFM modulation method at the similar power level.

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TABLE III
COMPARISON BETWEEN THE PROPOSED CONVERTER AND OTHER CONVERTERS

Reference	[3]	[16]	[17]	[6]	Proposed
Modulation	PFM	PFM	PFM+PWM	PWM	PWM
Control loop			YES		No
Sensing circuit			YES		No
V_{in} (V)		400		380-420	400
V_o (V)	250-400	100-420	100-500	250-420	250-400
f_s (kHz)	75-300	102-236	70-150	50	100
η@ Full load	98.5%	95.21%	95.38%	98%@CC 97%@CV	96.7%
Rated power	3.2kW	1.5 kW	1.5 kW	3.3 kW	1 kW

V. CONCLUSION

In this letter, a novel *LCL* resonant converter with natural CC-CV characteristics is proposed for high voltage PEV charging applications. The circuit configuration, theoretical analysis and design considerations are described in detail. The experimental results from an 1 kW prototype with 400 V input and 250–400 V output show that the circuit achieves the full load efficiency of 96.7% and higher than 96% for the CC mode. Thus, it is not difficult to summarize the distinctive features of the proposed converter as follows.

1) Inherent CC-CV characteristics can eliminate the sensing and control circuit. Thus, the reliability and cost can be significantly improved.
2) Fixed resonant frequency operation can greatly reduce the circulating energy.
3) The *LCL* resonant tank can maintain ZVS operation for primary side switches and ZCS switching for secondary diodes over whole load range. Thus, the high efficiency can be obtained.

REFERENCES

[1] Z. Liu, B. Li, F. C. Lee, and Q. Li, "High-efficiency high-density critical mode rectifier/inverter for WBG-device-based on-board charger," *IEEE Trans. Ind. Electron.*, vol. 64, no. 11, pp. 9114–9123, Nov. 2017.

[2] C.-Y. Oh, D.-H. Kim, D.-G. Woo, W.-Y. Sung, Y.-S. Kim, and B.-K. Lee, "A high-efficient nonisolated single-stage on-board battery charger for electric vehicles," *IEEE Trans. Power Electron.*, vol. 28, no. 12, pp. 5746–5757, Dec. 2013.

[3] L. A. D. Ta, N. D. Dao, and D.-C. Lee, "High-efficiency hybrid LLC resonant converter for on-board chargers of plug-in electric vehicles," *IEEE Trans. Power Electron.*, vol. 35, no. 8, pp. 8324–8334, Aug. 2020.

[4] M. Escudero, D. Meneses, N. Rodriguez, and D. P. Morales, "Modulation scheme for the bidirectional operation of the phase-shift full-bridge power converter," *IEEE Trans. Power Electron.*, vol. 35, no. 2, pp. 1377–1391, Feb. 2020.

[5] C. Li, H. Wang, and M. Shang, "A five-switch bridge based reconfigurable LLC converter for deeply depleted PEV charging applications," *IEEE Trans. Power Electron.*, vol. 34, no. 5, pp. 4031–4035, May 2019.

[6] H.-N. Vu and W. Choi, "A novel dual full-bridge LLC resonant converter for CC and CV charges of batteries for electric vehicles," *IEEE Trans. Ind. Electron.*, vol. 65, no. 3, pp. 2212–2225, Mar. 2018.

[7] J.-Y. Lee and H.-J. Chae, "6.6-kW onboard charger design using DCM PFC converter with harmonic modulation technique and two-stage DC/DC converter," *IEEE Trans. Ind. Electron.*, vol. 61, no. 3, pp. 1243–1252, Mar. 2014.

[8] W. Yu, J.-S. Lai, W.-H. Lai, and H. Wan, "Hybrid resonant and PWM converter with high efficiency and full soft-switching range," *IEEE Trans. Power Electron.*, vol. 27, no. 12, pp. 4925–4933, Dec. 2012.

[9] M. Borage, S. Tiwari, and S. Kotaiah, "LCL-T resonant converter with clamp diodes: A novel constant-current power supply with inherent constant-voltage limit," *IEEE Trans. Ind. Electron.*, vol. 54, no. 2, pp. 741–746, Apr. 2007.

[10] S. Wang, Y. Liu, and X. Wang, "Resonant converter for battery charging applications with CC/CV output profiles," *IEEE Access*, vol. 8, pp. 54879–54886, 2020.

[11] Z. Huang, S.-C. Wong, and C. K. Tse, "Design of a single-stage inductive-power-transfer converter for efficient EV battery charging," *IEEE Trans. Veh. Technol.*, vol. 66, no. 7, pp. 5808–5821, Jul. 2017.

[12] J. Deng, S. Li, S. Hu, C. C. Mi, and R. Ma, "Design methodology of LLC resonant converters for electric vehicle battery chargers," *IEEE Trans. Veh. Technol.*, vol. 63, no. 4, pp. 1581–1592, May 2014.

[13] X. Qu, H. Han, S.-C. Wong, C. K. Tse, and W. Chen, "Hybrid IPT topologies with constant current or constant voltage output for battery charging applications," *IEEE Trans. Power Electron.*, vol. 30, no. 11, pp. 6329–6337, Nov. 2015.

[14] Y. Chen, B. Yang, Z. Kou, Z. He, G. Cao, and R. Mai, "Hybrid and reconfigurable IPT systems with high-misalignment tolerance for constant-current and constant-voltage battery charging," *IEEE Trans. Power Electron.*, vol. 33, no. 10, pp. 8259–8269, Oct. 2018.

[15] U. Kundu, K. Yenduri, and P. Sensarma, "Accurate ZVS analysis for magnetic design and efficiency improvement of full-bridge LLC resonant converter," *IEEE Trans. Power Electron.*, vol. 32, no. 3, pp. 1703–1706, Mar. 2017.

[16] M. I. Shahzad, S. Iqbal, and S. Taib, "A wide output range HB-2LLC resonant converter with hybrid rectifier for PEV battery charging," *IEEE Trans. Transp. Electrif.*, vol. 3, no. 2, pp. 520–531, Jun. 2017.

[17] H. Wu, Y. Li, and Y. Xing, "LLC resonant converter with semiactive variable-structure rectifier (SA-VSR) for wide output voltage range application," *IEEE Trans. Power Electron.*, vol. 31, no. 5, pp. 3389–3394, May 2016.

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